

Varnan Rathod

(+91) 915-636-6939 | varnanrathod6@gmail.com | linkedin.com/in/varnan-rathod | github.com/varnan6

EDUCATION

Shri Ramdeobaba College of Engineering and Management

B.Tech. in Computer Science and Engineering (Honors in AI/ML): 9.25 CGPA

Nagpur, MH

Aug 2023 – May 2027

School of Scholars, Wanadongri

HSC: 92%

Nagpur, MH

Jul 2021 – May 2023

TECHNICAL SKILLS

Languages: Python, Java, Javascript, SQL, C

Frameworks: TensorFlow, Keras, PyTorch

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, LaTeX

Libraries: NumPy, Pandas, Seaborn, Matplotlib, Scikit-Learn, SciPy, XGBoost, SHAP

Databases: PostgreSQL, MongoDB

RESEARCH EXPERIENCE

Research Intern

Visvesvaraya National Institute of Technology (VNIT)

Dec 2025 – Present

Nagpur, MH (part time)

- Working under Dr. Praveen Kumar to design and optimize models related with vision and detection.

Undergraduate Researcher

Ramdeobaba University

Aug 2025 – Oct 2025

Nagpur, MH (part time)

Mentor: Dr. Khushboo Khurana

- Conducted a comparative study of Transformer-based object detection models, including DiffusionDet and DEIM.
- Analyzed architectural trade-offs and scene understanding mechanisms in DETR-style detectors.
- Used a curated dataset for a particular use case and benchmarked models using mAP across various object scales and confidence thresholds.
- Fine-tuned selected models, achieving state-of-the-art (SOTA) bounding box detection results on the dataset.

Research & Development Intern

Ramdeobaba University

May 2025 – Feb 2026

Nagpur, MH (Part-time)

Project: Development of Coal Quality Exploration Technique using CNNs and Hyperspectral Imaging

Sponsored by: Ministry of Coal, Government of India

Guide: Principal & Prof. M. B. Chandak | **Coordinator:** Dr. Abhijeet Raipurkar

- Implemented a CNN-based regression pipeline for coal quality assessment using hyperspectral image datacubes.
- Constructed and preprocessed a novel hyperspectral dataset acquired from a newly commissioned imaging sensor.
- Analyzed latent correlations across spectral bands to inform model adaptation and optimization.
- Fine-tuned a pretrained CNN architecture on a dataset approximately 3% the size of the original training corpus.
- Achieved a 72% reduction in MAE relative to baseline.

PUBLICATIONS & ACADEMIC SERVICE

- [Paper] Utilizing DETR and Diffusion-Based Models for Underwater Debris Detection and Recognition** - Accepted at *IEEE Recent Advances in Electronics, Communication, Computing, Automation, and Power (RECCAP) 2026*.
- [Reviewer]** — *IEEE Guwahati Subsection Conference (GCON) 2026*

TESTS & SCORES

- GATE CS 2026:** AIR 2472, Score: 625
- JEE Advanced:** Secured an All India Rank (AIR) of **22704**
- JEE Main (B.Tech):** Achieved a percentile of **95.72%ile**
- JEE Main (B.Arch):** Attained **99.43%ile** with an All India Rank (AIR) of **362**
- MHT-CET:** Secured a percentile of **98.77%ile**
- Bebras Computing Challenge 2022:** Ranked **9th** nationally out of 6,369 participants

RELEVANT COURSES

Course Code	Course Name	Grade
CATH 6100	Generative Adversarial Networks (ongoing)	–
CATH 5100	AI for Vision and Navigation	AA
CATH 4100	Distributed Computing Frameworks	BB
CATH 3100	Computational Statistics For Data Science	AB
CST 6003	Machine Learning (ongoing)	–
CST 5003	Artificial Intelligence	AA
CST 4002	Design and Analysis of Algorithms	BB
CST 3001	Data Structures	AB
MAT 4001	Linear Algebra	AA
MAT 3001	Probability and Statistics	AA
MAT 2001	Discrete Mathematics	AA
MAT 1001	Calculus	AA

ACHIEVEMENTS & ROLES

- **Finalist:** Datathon, ARTIMAS'25, PCCOE Pune
- **Grand Finalist:** Smart India Hackathon (SIH) 2024
- **LeetCode Rating:** 1566 (460+ problems solved)
- **CodeChef Rating:** 1496 (2, 700+ problems solved)
- **Executive Tech Member** — GFG RBU Chapter
- **Academic Mentor (2024-2025 & 2025-2026)** — RBU
- **Web Development Member** — Technical Club, RBU
- **City Rank 2:** CSIR NEERI Science Quiz 2023